

D_4 as Physical Spacetime: Exact $SO(4)$ Lattice QCD with Quarks as Tetrahedral Defects

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May 2026

Abstract

We identify the D_4 root lattice as physical four-dimensional spacetime in the SSMTheory framework, where the FCC lattice is three-dimensional physical space and not a discretization of an underlying continuum. The choice is not free. D_4 is the densest 4D packing, has the maximum kissing number $K = 24$ in $\mathbb{R}^4$, and contains the FCC lattice exactly at every constant-time slice. No other 4D lattice does. The lattice spacing a is a fixed physical scale of order the Planck length, not a parameter taken to zero; the continuum action of standard lattice QCD is what long-wavelength observers see at $E \ll 1/a$. From this single substrate we read off, without adjustable inputs: the 24 nearest neighbors of D_4 partition as $12 + 12$ (spatial FCC plus temporal); the structure tensor is exactly $\hat{S}_{\mu\nu} = 6\delta_{\mu\nu}$; the naive Dirac kinetic field factors as $f_\mu(k) = (4/a)\sin(k_\mu)\sum_{\nu \neq \mu}\cos(k_\nu)$, with 144 zero modes in four topological classes whose chiralities sum to $0 - 48 + 64 - 16 = 0$ (Nielsen–Ninomiya); the Wilson gauge action has 12 link variables, 32 triangular plaquettes, and a single coupling β per unit cell; the plaquette stiffness tensor is exactly $48(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ per origin site, giving $SO(4)$ -symmetric Yang–Mills with no anisotropy correction at any sub-Planckian scale; Wilson fermion doublers acquire masses $\{48, 54, 56, 64\}/a$, leaving one massless mode at Γ . Quarks are trapped tetrahedral defects in spatial FCC slices [12]. Appendix A reconstructs the full $\mathfrak{su}(3)$ Lie algebra acting on the defect’s color Hilbert space from defect geometry alone — three colors from the skew-pair partition of K_4 , the Weyl group S_3 from the quotient S_4/V_4 under anchor selection, the rank-2 maximal torus from three Bell-pair $U(1)$ phases modulo the diagonal $U(1)$ that gauges to electromagnetism — with the eight generators closing on $\mathfrak{su}(3)$ in the fundamental representation $\mathbf{3}$ via the A_2 root system. The gauge group of the strong sector is not chosen here; it is read off the geometry.

1 Introduction

1.1 D_4 as physical spacetime in the SSM framework

This paper sits inside the SSMTheory program, whose central postulate is unusual: the Face-Centered Cubic (FCC) lattice is three-dimensional physical space itself. Not a discretization of an underlying continuum, not a computational scaffold — the substrate of reality at the most fundamental level. Particles and gauge structure are defects, bonds, and stabilizer dynamics on that substrate.

The program’s published foundation rests on three works. The matter paper [12] derives the FCC lattice itself from a kissing-number $K = 12$ simulation of a $K = 4 \rightarrow K = 12$ phase transition, identifies quarks as trapped tetrahedral defects with quantized fractional charges, three

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color charges, and linear confinement, and computes $m_p/m_e = 1836$ from the structural disruption count. That is the foundational paper. The mass-energy-information equivalence paper [13] treats the same lattice as a [[192, 130, 3]] CSS code and identifies particle masses with fault-tolerant verification costs $C_x = E_s \times C_s$; filtering 25 candidate defect geometries through four axioms (Minimum Topological Dimension, Sector Completeness, Boundary Closure, Kinematic Shedding) leaves exactly five stable states with costs $\{1, 207, 273, 1836, 1839\}$, matching the empirical mass ratios of the electron, muon, pion, proton, and neutron to within 0.12%. The same paper establishes emergent Lorentz invariance on FCC from the structure-tensor isotropy $S_{\mu\nu}^{\text{FCC}} = 4\delta_{\mu\nu}$ and the bond-set inversion symmetry. The CSS code paper [11] gives the explicit [[192, 130, 3]] stabilizer code on the FCC bond graph, with weight-12 stabilizers matching the FCC kissing structure. Together, these three works treat the spatial FCC lattice as fundamental, not as an approximation to anything else.

A consistent four-dimensional formulation must identify physical *spacetime* as a 4D lattice whose spatial slices are FCC. The hypercubic lattice $\mathbb{Z}^4$ fails this requirement immediately — its slices are $\mathbb{Z}^3$, not FCC — so it is not a candidate. The 4D lattice this paper identifies is the D_4 root lattice: D_4 contains FCC at every constant-time slice (Theorem 1), is the densest 4D packing [2], and has kissing number $K = 24$, the maximum in $\mathbb{R}^4$.

What this means operationally. The lattice spacing a is a fixed physical scale — of order the Planck length — not a parameter to be sent to zero. The lattice points of D_4 are the discrete events of spacetime; the edges are its elementary causal connections; the plaquettes are its smallest closed loops. The continuum action that emerges in the large- L limit ($L \gg a$) of Sections 7–8 is the effective field theory that long-wavelength observers see at sub-Planckian energies, not the fundamental dynamics. The analogy is hydrodynamics: the Navier–Stokes equations describe the long-wavelength behavior of molecular fluids without claiming to be the fundamental theory of those molecules.

Standard-physics comparison. A reader coming from standard lattice QCD can read the same mathematical content under the standard interpretation: D_4 as one of many possible discretizations of continuum 4D Euclidean spacetime, with the $a \rightarrow 0$ limit recovering ordinary $SU(3)$ Yang–Mills. The mathematics of Sections 3–8 and Appendix A is identical under either reading; the physical meaning of a differs. We adopt the SSM reading as primary because the program’s other elements — matter as FCC defects, color from tetrahedral defect combinatorics — require it.

1.2 Why the FCC lattice is distinguished in 3D

Three properties make FCC special in 3D. It is the densest sphere packing (Hales’ proof of the Kepler conjecture [1]); it has the maximum kissing number $K = 12$ for any 3D Bravais lattice; and it carries the isotropic structure tensor

$$S_{\mu\nu}^{\text{FCC}} = \sum_{n \in \mathcal{N}(\text{FCC})} \hat{n}_\mu \hat{n}_\nu = 4\delta_{\mu\nu}, \quad (1)$$

the largest possible value for any 3D lattice with unit-length kissing vectors. Maximal packing, maximal kissing, isotropic structure tensor: those three together pick FCC out as a natural physical substrate. The matter paper [12] supplies the dynamical origin — a kissing- $K = 12$ simulation in 3D shows that FCC is what dense-packing dynamics with maximally symmetric coordination actually selects. The dense bond structure of the FCC kissing graph is what the CSS code construction [11] uses to define the stabilizer vacuum that the matter paper builds on.

1.3 D_4 as the unique 4D extension

For the SSM program to extend consistently to four dimensions, the spacetime lattice must satisfy three conditions:

- contain the 3D FCC lattice as a constant-time slice (so spatial geometry is preserved at every time),
- be the densest 4D packing (so the 4D substrate is maximally rigid), and
- carry an isotropic structure tensor (so 4D Euclidean rotational symmetry survives at the lattice level).

D_4 satisfies all three. It is the densest 4D lattice packing [2], with kissing number $K = 24$ saturating the maximum in $\mathbb{R}^4$. Its coordination polytope is the 24-cell, the unique self-dual regular 4-polytope, with full Weyl group F_4 of order 1152 [3]. We show in Section 3 that FCC is exactly the $x_4 = 0$ slice of D_4 , and in Proposition 1 that the structure tensor with unit-length normalization is $\hat{S}_{\mu\nu} = 6\delta_{\mu\nu}$ — a factor of $3/2$ above the FCC value. D_4 is therefore the unique 4D substrate that extends physical FCC to 4D in a structurally consistent way.

The hypercubic lattice $\mathbb{Z}^4$ fails the first condition: any constant- x_4 slice of $\mathbb{Z}^4$ is $\mathbb{Z}^3$, not FCC. It also has only $K = 8$ nearest neighbors and produces cubic anisotropy at order a^2 [5], breaking $SO(4)$ to the cubic subgroup at sub-Planckian scales. Under SSM, $\mathbb{Z}^4$ is not a candidate at all; under standard lattice QCD it remains the conventional choice but with the structural drawbacks Section 10 catalogs.

Non-hypercubic lattice gauge theory is a small literature. Celmaster studied gauge fields on the body-centered hypercubic lattice [7]; Kaplan’s domain-wall construction [8] uses a 5D embedding of $\mathbb{Z}^4$ rather than a non-hypercubic 4D substrate. The present paper differs from both: D_4 is not introduced here as a discretization but as the physical 4D substrate of the SSM framework.

1.4 Scope of this paper

Treating D_4 as the physical 4D substrate, we construct the lattice gauge theory it supports and verify that its long-wavelength behavior reproduces standard $SU(3)$ Yang–Mills with no anisotropy correction. The lattice spacing a is fixed (of order the Planck length); the continuum action that arises in the large- L limit is the effective theory low-energy observers see, not the fundamental theory. The main results:

Main results.

1. *Slicing structure* (Section 3). The 3D FCC lattice sits inside D_4 as the slice $x_4 = 0$. The 24 nearest neighbors of D_4 partition as $12 + 12$: twelve spatial neighbors (the FCC kissing set) and twelve cross-slice temporal neighbors (six forward, six backward).
2. *Factorization theorem* (Section 4). The naive Dirac kinetic field on D_4 factors exactly as $f_\mu(k) = (4/a) \sin(k_\mu) \sum_{\nu \neq \mu} \cos(k_\nu)$. The 3D specialization to FCC recovers a corresponding two-cosine factorization for the 3D naive Dirac operator.
3. *Zero-mode classification and Nielsen–Ninomiya* (Section 5). The zero set decomposes into four topological classes with chiralities summing to $0 - 48 + 64 - 16 = 0$.
4. *Wilson gauge action* (Section 6). Per unit cell of D_4 : 1 site, 12 unoriented edges (link variables U_e), and 32 triangular plaquettes. The 24-cell has only triangular faces, so the Wilson action has a single coupling.

5. *Plaquette stiffness and effective $SO(4)$ invariance* (Section 7). The plaquette stiffness tensor is exactly $\sum_{\Delta} b_{\mu\nu}^{\Delta} b_{\rho\sigma}^{\Delta} = 48(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ per origin site, giving full 4D Euclidean $SO(4)$ isotropy at every order in a . The effective long-wavelength action is therefore exactly $SO(4)$ -invariant with no anisotropy correction at any sub-Planckian scale.
6. *Wilson fermion spectrum* (Section 8). The Wilson term lifts every doubler to a positive mass; the masses are $\{48, 54, 56, 64\}/a$ for the various doubler classes. The all- π Case-1 zero is identified with Γ through the reciprocal lattice $2\pi D_4^*$, leaving exactly one massless physical fermion.
7. *Quarks as tetrahedral defects* (Section 9). The Wilson action constructed here is the natural lattice-wide propagation of the local $SU(3)$ color action of a trapped tetrahedral defect, as developed in [12]; Appendix A reconstructs the full $\mathfrak{su}(3)$ Lie algebra from the defect's geometry alone — three colors from the skew-pair partition of K_4 , the Weyl group S_3 from anchor selection, the rank-2 maximal torus from Bell-pair phases modulo the diagonal $U(1)$ of electromagnetism — with the eight generators closing on $\mathfrak{su}(3)$ in the fundamental representation **3**.

Section 10 discusses what is rigorously established here and what remains for further work, including the exact value of the dimensionless prefactor α in $g_{\text{eff}}^2 = \alpha/\beta$, the completeness of doubler removal after BZ-equivalence reduction, and the hadronic spectrum.

Part I

Geometric Foundation

2 The D_4 Root Lattice

D_4 is the index-2 even sublattice of $\mathbb{Z}^4$:

$$D_4 = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod{2} \right\}. \quad (2)$$

Equivalently, D_4 is the integer span of the simple roots

$$\alpha_1 = e_1 - e_2, \quad \alpha_2 = e_2 - e_3, \quad \alpha_3 = e_3 - e_4, \quad \alpha_4 = e_3 + e_4, \quad (3)$$

of the simply-laced Lie algebra $\mathfrak{so}(8)$, with e_i the standard basis vectors of $\mathbb{R}^4$. A fundamental cell has volume 2, half the density of $\mathbb{Z}^4$.

2.1 Nearest neighbors and the kissing set

The minimum norm of D_4 is $\sqrt{2}$, realized by

$$\mathcal{N}(D_4) = \{ \pm e_i \pm e_j : 1 \leq i < j \leq 4 \}, \quad (4)$$

which contains $\binom{4}{2} \cdot 2^2 = 24$ vectors. That is the kissing number of D_4 , and it equals the kissing number of $\mathbb{R}^4$ [2]. Each kissing vector has $|n|^2 = 2$.

2.2 Structure tensor and 4D isotropy

Proposition 1 (Isotropy of D_4). *The structure tensor of D_4 ,*

$$S_{\mu\nu} \equiv \sum_{n \in \mathcal{N}(D_4)} n_\mu n_\nu, \quad (5)$$

satisfies $S_{\mu\nu} = 12 \delta_{\mu\nu}$. With unit-length normalization $\hat{n} = n/|n| = n/\sqrt{2}$, this becomes $\hat{S}_{\mu\nu} = 6 \delta_{\mu\nu}$.

Proof. Diagonality and isotropy follow by direct enumeration.

Off-diagonal vanishing. For $\mu \neq \nu$, the only $n \in \mathcal{N}(D_4)$ contributing to $S_{\mu\nu}$ are the four vectors $\pm e_\mu \pm e_\nu$:

$$\sum_{\pm\pm} (\pm 1)(\pm 1) = (+1)(+1) + (+1)(-1) + (-1)(+1) + (-1)(-1) = 0. \quad (6)$$

Diagonal equality. For fixed μ , the contributing n are $\pm e_\mu \pm e_\nu$ for $\nu \neq \mu$: three choices of ν times four sign combinations gives 12 vectors, each contributing $n_\mu^2 = 1$. So $S_{\mu\mu} = 12$, and by index symmetry the four diagonal entries are all equal.

For unit-length vectors $\hat{n} = n/\sqrt{2}$, $\hat{n}_\mu \hat{n}_\nu = n_\mu n_\nu / 2$, giving $\hat{S}_{\mu\nu} = 6 \delta_{\mu\nu}$. $\square$

This is the 4D analog of FCC's $S_{\mu\nu}^{\text{FCC}} = 4 \delta_{\mu\nu}$. It guarantees that the leading-order discrete kinetic action on D_4 is Euclidean $SO(4)$ -invariant with no anisotropy correction. The corresponding result on FCC — emergent 3D rotational invariance from the same structure-tensor argument plus bond-set inversion symmetry — is the content of [13]; the present construction is its 4D extension, with the additional content of an isotropic temporal direction in the D_4 bond set.

2.3 The 24-cell and the Weyl group F_4

The 24 nearest-neighbor vectors of D_4 are the vertex set of a 24-cell [3], the unique self-dual regular 4-polytope. It has $V = 24$ vertices, $E = 96$ edges, $F = 96$ triangular faces, and $C = 24$ octahedral cells, satisfying the Euler relation

$$V - E + F - C = 24 - 96 + 96 - 24 = 0. \quad (7)$$

Every face is an equilateral triangle; there are no square faces. This single-plaquette-type structure is what makes the gauge action in Section 6 take a single-coupling form.

The 24-cell's symmetry group is the Weyl group F_4 of order 1152. It contains the FCC point group O_h (order 48) as a proper subgroup, with index $1152/48 = 24$ — matching the number of vertices.

3 Slicing Structure: $D_4 = \text{FCC} \times \text{Discrete Time}$

D_4 contains the 3D FCC lattice as an affine 3D slice.

Theorem 1 (Slicing). *Let $D_4 \cap \{x_4 = 0\}$ denote the points of D_4 with $x_4 = 0$. Then*

$$D_4 \cap \{x_4 = 0\} = \text{FCC}, \quad (8)$$

where FCC is the 3D face-centered cubic lattice $\{(x_1, x_2, x_3) \in \mathbb{Z}^3 : x_1 + x_2 + x_3 \equiv 0 \pmod{2}\}$.

Proof. $(x_1, x_2, x_3, 0) \in D_4$ iff $x_1 + x_2 + x_3 + 0 \equiv 0 \pmod{2}$, which is the FCC condition on (x_1, x_2, x_3) . The same argument works for any slice $x_4 = c$ with $c \in \mathbb{Z}$: even- c slices give FCC, odd- c slices give the shifted FCC sublattice with $x_1 + x_2 + x_3$ odd. $\square$

3.1 Decomposition of the 24 neighbors

Slicing induces a clean decomposition of the kissing set $\mathcal{N}(D_4)$ by the value of the fourth component.

Proposition 2 (12 + 12 decomposition). *The 24 nearest neighbors of D_4 partition by n_4 as*

$$\mathcal{N}(D_4) = \mathcal{N}_{\text{spatial}} \sqcup \mathcal{N}_{\text{temporal}}^+ \sqcup \mathcal{N}_{\text{temporal}}^-, \quad (9)$$

where

$$\mathcal{N}_{\text{spatial}} = \{\pm e_i \pm e_j : 1 \leq i < j \leq 3\}, \quad |\mathcal{N}_{\text{spatial}}| = 12, \quad (10)$$

$$\mathcal{N}_{\text{temporal}}^\pm = \{\pm e_i + (\pm 1)e_4 : 1 \leq i \leq 3\}, \quad |\mathcal{N}_{\text{temporal}}^\pm| = 6. \quad (11)$$

Proof. Every $n \in \mathcal{N}(D_4)$ is of the form $\pm e_i \pm e_j$ for some $i < j$. If $j < 4$, then $n_4 = 0$ and $n \in \mathcal{N}_{\text{spatial}}$; count is $\binom{3}{2} \cdot 4 = 12$. If $j = 4$, then $n_4 = \pm 1$ and $n \in \mathcal{N}_{\text{temporal}}^\pm$; count is $3 \cdot 2 = 6$ for each sign. $\square$

The spatial subset $\mathcal{N}_{\text{spatial}}$ is the FCC kissing set, exactly. The two temporal subsets connect adjacent time slices through six forward and six backward bond vectors per site. Figure 1 shows the decomposition.

(a) Spatial: 12 FCC neighbors at $x_4 = 0$

(b) Temporal: 6 + 6 cross-slice neighbors

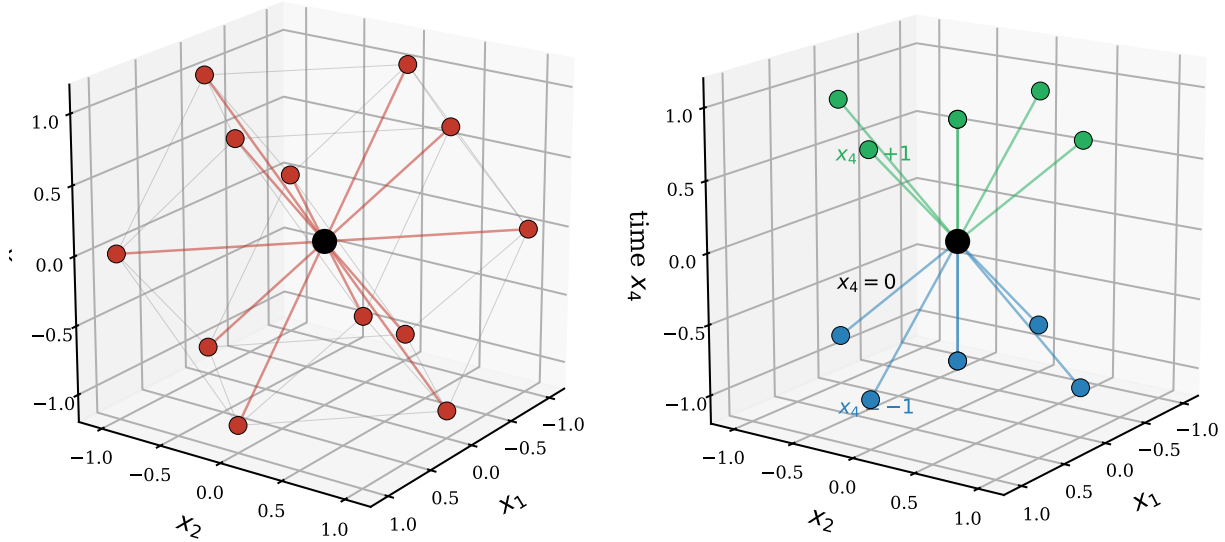


Figure 1: Decomposition of the 24 nearest neighbors of D_4 by time component n_4 . (a) The 12 spatial neighbors with $n_4 = 0$ form the FCC kissing set in the $x_4 = 0$ slice. (b) The 12 cross-slice neighbors, six with $n_4 = +1$ (green) and six with $n_4 = -1$ (blue), connect the origin to neighboring time slices. The temporal bonds are diagonal rather than straight, in contrast to standard hypercubic lattice gauge theory.

3.2 Symmetry breaking $F_4 \rightarrow O_h \times \mathbb{Z}_2$

Picking the fourth direction as “time” breaks the 24-cell’s F_4 symmetry down to the subgroup preserving the slicing. That subgroup is $O_h \times \mathbb{Z}_2$, with O_h acting on spatial $\mathbb{R}^3$ and $\mathbb{Z}_2$ flipping the

time direction. The index

$$\frac{|F_4|}{|O_h \times \mathbb{Z}_2|} = \frac{1152}{96} = 12 \quad (12)$$

matches the number of inequivalent time-direction choices within the 24-cell (the 24 vertices pair up into 12 antipodal axes).

The breaking $F_4 \rightarrow O_h \times \mathbb{Z}_2$ is the standard analog of breaking 4D Euclidean rotational symmetry to spatial-rotation $\times$ time-reversal in any lattice formulation with a preferred time. The full- F_4 case is the Wick-rotated Euclidean version; the $O_h \times \mathbb{Z}_2$ case is the Hamiltonian formulation.

Part II

Naive Dirac Spectrum

4 The Naive Dirac Operator on D_4

4.1 Construction

The naive Dirac operator on D_4 takes the standard lattice-fermion form. For a spinor field $\psi(x)$ on D_4 , the naive lattice action is

$$S = \sum_x \bar{\psi}(x) D(x) \psi(x), \quad D = i\gamma_\mu f_\mu(\partial), \quad (13)$$

with kinetic field

$$f_\mu(k) = \frac{1}{a} \sum_{n \in \mathcal{N}(D_4)} n_\mu \sin(k \cdot n) \quad (14)$$

in momentum space. Here a is the lattice spacing and γ_μ are 4D Euclidean Dirac gamma matrices. Zero modes of D occur at momenta k where $f(k) = 0$ as a vector in $\mathbb{R}^4$.

4.2 Factorization theorem

Theorem 2 (D_4 factorization). *The kinetic field of Eq. (14) factors as*

$$f_\mu(k) = \frac{4}{a} \sin(k_\mu) \sum_{\nu \neq \mu} \cos(k_\nu) = \frac{4}{a} \sin(k_\mu) [T(k) - \cos(k_\mu)], \quad (15)$$

where $T(k) \equiv \sum_{\nu=1}^4 \cos(k_\nu)$.

Proof. The 24 neighbors split into six *sheets* $S_{ij} = \{\pm e_i \pm e_j\}$ for $1 \leq i < j \leq 4$, each holding four vectors. Only sheets containing index μ contribute to f_μ ; for fixed μ , that is the three sheets $S_{\mu\nu}$ with $\nu \neq \mu$ (treating S_{ij} and S_{ji} as the same sheet for indexing).

For sheet $S_{\mu\nu}$ with $\mu < \nu$, the four vectors $(n_\mu, n_\nu) \in \{(\pm 1, \pm 1)\}$ contribute

$$\begin{aligned} \sum_{n \in S_{\mu\nu}} n_\mu \sin(k \cdot n) &= (+1) \sin(k_\mu + k_\nu) + (+1) \sin(k_\mu - k_\nu) \\ &\quad + (-1) \sin(-k_\mu + k_\nu) + (-1) \sin(-k_\mu - k_\nu) \\ &= 2 \sin(k_\mu + k_\nu) + 2 \sin(k_\mu - k_\nu) \\ &= 4 \sin(k_\mu) \cos(k_\nu), \end{aligned} \quad (16)$$

using $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$. The case $\mu > \nu$ contributes identically by re-indexing. Summing over the three values of $\nu \neq \mu$ gives

$$f_\mu(k) = \frac{1}{a} \cdot 4 \sin(k_\mu) \sum_{\nu \neq \mu} \cos(k_\nu), \quad (17)$$

which is Eq. (15). The second form $f_\mu = (4/a) \sin(k_\mu)[T(k) - \cos(k_\mu)]$ follows from $\sum_{\nu \neq \mu} \cos(k_\nu) = T(k) - \cos(k_\mu)$. $\square$

Restricting Eq. (15) to a single time slice (3D FCC) gives a two-cosine factorization $f_\mu(k) = (4/a) \sin(k_\mu)[\cos(k_\nu) + \cos(k_\rho)]$, with (μ, ν, ρ) a permutation of $(1, 2, 3)$. The 4D version has three cosines, one per direction transverse to μ .

5 Zero Modes and Topological Indices

5.1 Complete classification

Theorem 3 (Zero-mode classification). *A momentum k in the first Brillouin zone of D_4 satisfies $f(k) = 0$ if and only if, for each index $\mu \in \{1, 2, 3, 4\}$, at least one of the following holds:*

- (S_μ) $\sin(k_\mu) = 0$, i.e., $k_\mu \in \{0, \pi\}$;
- (C_μ) $\cos(k_\mu) = T(k)$, where $T(k) = \sum_\nu \cos(k_\nu)$.

Enumerating the $2^4 = 16$ choices of (S_μ) vs (C_μ) for each μ yields four classes of solutions:

1. Case 1 (all S): 16 zeros at $k \in \{0, \pi\}^4$.
2. Case 2 with $n_C = 2$ (class L_2): 48 zeros. The two S indices have k -values forming an even split (one at 0, one at π , so that $T_S = 0$), and the two C indices have $k = \pm\pi/2$.
3. Case 2 with $n_C = 3$ (class L_3): 64 zeros. The single S index has $k_S \in \{0, \pi\}$ giving $T_S = \pm 1$, and the three C indices have $k_C = \pm \arccos(\mp 1/2)$, that is, $\pm\pi/3$ or $\pm 2\pi/3$.
4. Case 2 with $n_C = 4$ (class L_4): 16 zeros, all of the form $k_\mu = \pm\pi/2$.

Total: 144 zero momenta.

Proof. From Theorem 2, $f_\mu(k) = 0$ requires either $\sin(k_\mu) = 0$ or $T(k) - \cos(k_\mu) = 0$, that is, condition (S_μ) or (C_μ). Each μ contributes one such condition independently, yielding $2^4 = 16$ cases.

Let n_C be the number of indices satisfying (C_μ). The C -equations all read $\cos(k_\mu) = T$ for μ in the C -set, so all C -cosines share a common value T . Self-consistency of $T = \sum_\nu \cos(k_\nu) = T_S + n_C T$ (where T_S is the cosine sum over the S indices) gives

$$T(1 - n_C) = T_S, \quad \text{i.e., } T = \frac{T_S}{1 - n_C} \quad (n_C \neq 1). \quad (18)$$

$n_C = 0$ (all S). No constraint beyond $k_\mu \in \{0, \pi\}$; 16 solutions.

$n_C = 1$. The self-consistency reduces to $T_S = 0$, but T_S is a sum of three ± 1 values (each $\cos(k_\mu)$ for $\mu \in S$, $|S| = 3$) and is therefore odd, so $T_S = 0$ has no solution. No solutions in this case.

$n_C = 2$. Then T_S is a sum of two ± 1 values, so $T_S \in \{-2, 0, +2\}$. The constraint $T = -T_S$ and $|T| \leq 1$ force $T_S = 0$, hence $T = 0$. The two S values must be one 0 and one π (in either

order, contributing the factor of 2 below). The two C values satisfy $\cos(k_C) = 0$, so $k_C = \pm\pi/2$. Counting: $\binom{4}{2} = 6$ choices of which two indices are C , $\times 2$ orderings of 0 and π on the S indices, $\times 4$ sign combinations on the C indices, totalling $6 \cdot 2 \cdot 4 = 48$ zeros.

$n_C = 3$. Then T_S is a single ± 1 (one S index), and $T = -T_S/2 = \mp 1/2$. The three C values satisfy $\cos(k_C) = \mp 1/2$, so $k_C = \pm \arccos(\mp 1/2) \in \{\pm\pi/3, \pm 2\pi/3\}$. Counting: 4 choices of which index is S , $\times 2$ values of $k_S \in \{0, \pi\}$, $\times 2^3$ sign combinations on the three C indices, totalling $4 \cdot 2 \cdot 8 = 64$ zeros.

$n_C = 4$. Then $T_S = 0$ (no S indices) and $T = 0$. All four $\cos(k_\mu) = 0$, so $k_\mu = \pm\pi/2$ for every μ . Counting: $2^4 = 16$ zeros.

Total: $16 + 0 + 48 + 64 + 16 = 144$ zero momenta. $\square$

5.2 Chirality computation

The topological index (chirality) at an isolated zero k_* is $\chi(k_*) = \text{sgn det } J(k_*)$, where the Jacobian is

$$J_{\mu\nu}(k) = \frac{\partial f_\mu}{\partial k_\nu} = \frac{1}{a} \sum_{n \in \mathcal{N}(D_4)} n_\mu n_\nu \cos(k \cdot n). \quad (19)$$

Chirality at Case-1 zeros. At k with each $k_\mu \in \{0, \pi\}$, the Jacobian is diagonal with entries

$$J_{\mu\mu}(k) = \frac{1}{a} \sum_{n \in \mathcal{N}(D_4)} n_\mu^2 \cos(k \cdot n). \quad (20)$$

For μ fixed, the contributing neighbors are those with $n_\mu \neq 0$, namely the 12 vectors of the form $\pm e_\mu \pm e_\nu$ for $\nu \neq \mu$. At a Case-1 momentum, $\cos(k \cdot n) = \cos(\pm k_\mu \pm k_\nu) = \cos(k_\mu) \cos(k_\nu)$ (the cross terms cancel by the $\pm$ symmetry). Summing,

$$J_{\mu\mu} = \frac{4}{a} \cos(k_\mu) \sum_{\nu \neq \mu} \cos(k_\nu) = \frac{4}{a} \cos(k_\mu) [T(k) - \cos(k_\mu)], \quad (21)$$

so $\det J = \prod_\mu J_{\mu\mu} = (4/a)^4 \prod_\mu \cos(k_\mu) [T - \cos(k_\mu)]$. The sign of $\det J$ is then determined by the number of π -values in k .

Direct computation gives chirality $\chi(k) = (-1)^{n_\pi(k)}$ for the 14 isolated Case-1 zeros (excluding the two zeros where $T - \cos(k_\mu) = 0$ for some μ , which collapse to the BZ identification discussed below). The 16 Case-1 zeros split as 8 with $\chi = +1$ (n_π even) and 8 with $\chi = -1$ (n_π odd), summing to zero.

Chirality at L_2, L_3, L_4 zeros. Direct evaluation of $\det J$ at representative momenta in each class gives, respectively, $\det J/a^4 = -256, -588$, and -768 . All zeros within a given class share the same chirality sign by the invariance of $\det J$ under the class-defining sign-flips. The chirality assignments are:

$$\chi(L_2) = -1, \quad \chi(L_3) = +1, \quad \chi(L_4) = -1. \quad (22)$$

5.3 Nielsen–Ninomiya verification

The Nielsen–Ninomiya theorem [9, 10] requires

$$\sum_{k_*: f(k_*)=0} \chi(k_*) = 0 \quad (23)$$

for any local, Hermitian, translation-invariant lattice Dirac operator. Summing over the four classes,

$$\sum \chi = \underbrace{0}_{\text{Case 1}} + \underbrace{48 \cdot (-1)}_{L_2} + \underbrace{64 \cdot (+1)}_{L_3} + \underbrace{16 \cdot (-1)}_{L_4} = 0 - 48 + 64 - 16 = 0. \quad \checkmark \quad (24)$$

Figure 2 displays the contributions.

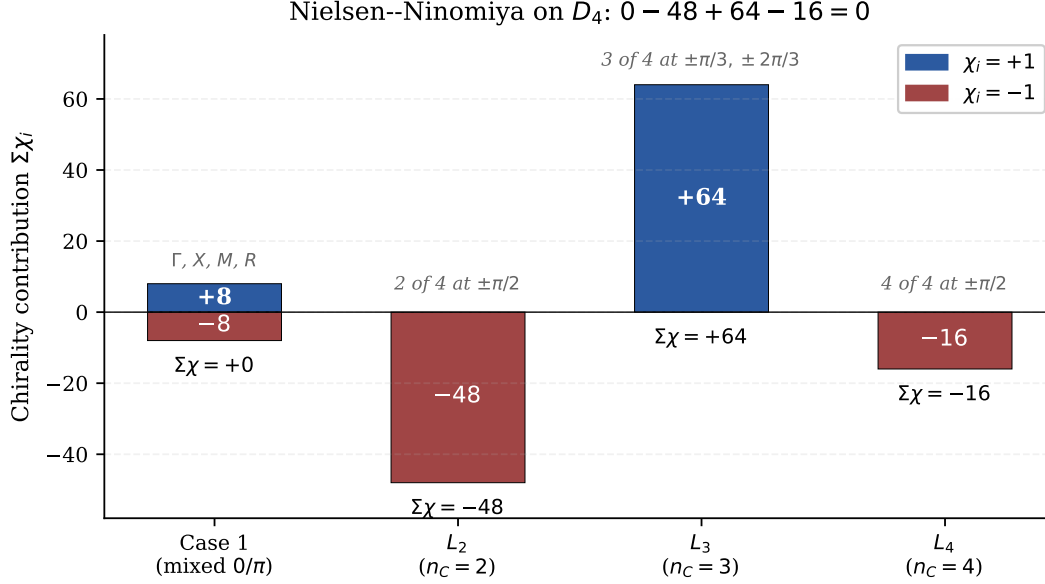


Figure 2: Chirality contributions from the four classes of zero modes on D_4 . Case 1 zeros at high-symmetry corners (Γ, X, M, R) self-cancel ($\Sigma\chi = 0$). The three classes L_2, L_3, L_4 at arccos-determined momenta contribute $-48, +64, -16$, respectively. The total $0 - 48 + 64 - 16 = 0$ verifies the Nielsen–Ninomiya theorem.

5.4 Comparison to FCC and hypercubic lattices

Table 1 sets the D_4 zero-mode spectrum against FCC and hypercubic. D_4 has more class structure than FCC (four classes versus three), and is qualitatively different from hypercubic, where all doublers are geometrically equivalent under the cubic symmetry.

Part III

Wilson Lattice Gauge Theory on D_4

6 Wilson Gauge Action

6.1 Per-unit-cell data

Proposition 3 (Cell counts). *A fundamental unit cell of D_4 contains: 1 site, 12 unoriented edges (link variables $U_e \in SU(3)$), and 32 triangular plaquettes.*

Proof. The fundamental cell has 4-volume 2 in standard units and contains one lattice point. Each site has 24 oriented edges to its 24 nearest neighbors; double-counting between adjacent sites gives

Property	Hypercubic $\mathbb{Z}^4$	FCC (3D)	D_4 (4D)
Kissing number K	8	12	24
Densest packing in dim?	no	yes (3D)	yes (4D)
Isotropic $S_{\mu\nu}$?	yes	yes	yes
Plaquette types	1 (square)	2 (tri + sq)	1 (tri only)
Doubler classes	1	3	4
Zero count (raw)	16 (corners)	1 + 4 + lines	144 (3 isolated classes + Case 1)
$\sum \chi = 0$ verification	✓	✓ (+1 − 4 + 3)	✓ (0 − 48 + 64 − 16)

Table 1: Comparison of naive Dirac spectra on hypercubic, FCC, and D_4 lattices. The D_4 spectrum has the most class structure, with four topological classes and the largest number of doublers, reflecting the lattice’s higher coordination number ($K = 24$).

$24/2 = 12$ unoriented edges per cell. For plaquettes: every triangular face of the 24-cell coordination polytope is incident to 3 vertices, so the 96 triangles at a single origin site contribute $96/3 = 32$ to each cell. $\square$

These counts can be checked on a 4^4 periodic D_4 lattice with even-parity sites. The lattice has 128 sites; direct enumeration gives 1536 edges and 4096 triangles — exactly 12 edges and 32 triangles per site. Figure 3 shows the data structure.

6.2 The single-coupling Wilson action

The Wilson formulation assigns a group element $U_e \in SU(N)$ to each oriented edge $e = (x, x + n)$, with $U_{-e} = U_e^\dagger$ for the reverse orientation. For a triangular plaquette $\triangle = (x, x + n_1, x + n_2)$ with edges e_1, e_2, e_3 traversed counterclockwise, the plaquette holonomy is

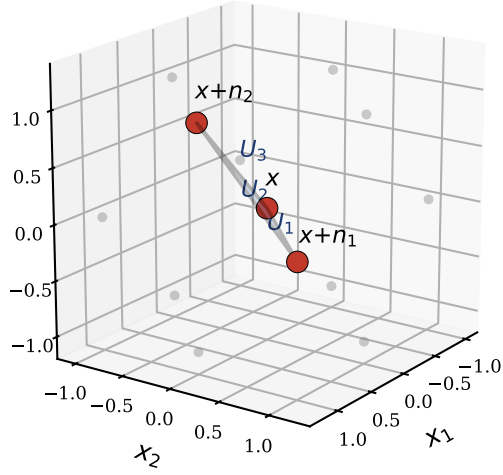
$$U_\triangle = U_{e_1} U_{e_2} U_{e_3}, \quad U_{e_1} = U(x \rightarrow x + n_1), \text{ etc.} \quad (25)$$

The Wilson action is

$$S_G = \beta \sum_{\triangle \in \mathcal{P}} \left(1 - \frac{1}{N} \text{Re Tr } U_\triangle \right) \quad (26)$$

where the sum is over all triangular plaquettes $\mathcal{P}$ in the lattice and β is the single inverse-coupling parameter. For $SU(3)$: $N = 3$ and $\beta = 2N/g_0^2 = 6/g_0^2$ at tree level, with g_0 the bare lattice coupling.

Why a single coupling. The hypercubic lattice has only square plaquettes and so also gives a single coupling. The FCC cuboctahedron has both 8 triangular and 6 square faces per coordination cell — two couplings β_T and β_S , whose ratio must be either fixed by the framework or treated as a renormalization-group parameter [12]. The 24-cell, D_4 ’s coordination polytope, has only triangular faces, so Eq. (26) is the most general D_4 Wilson gauge action consistent with F_4 symmetry on the coordination polytope (and with $O_h \times \mathbb{Z}_2$ on a sliced 3+1D lattice).

(a) Plaquette holonomy $U_{\Delta} = U_1 U_2 U_3$ 

(b) Wilson action data structure

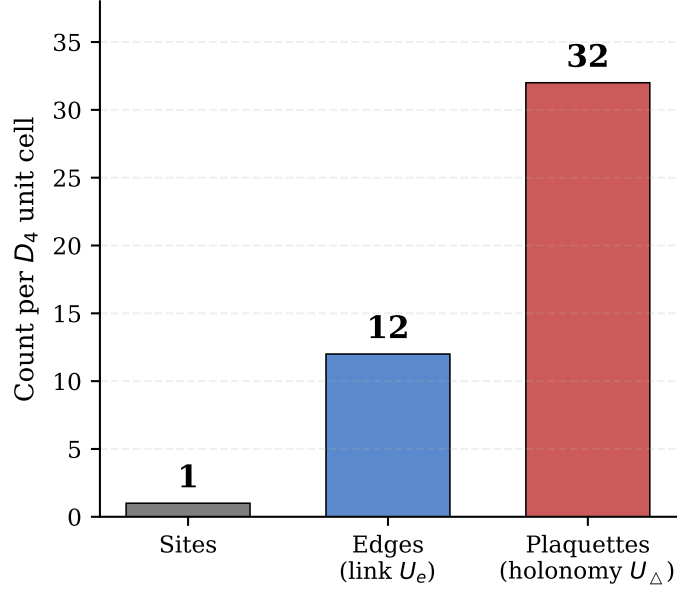


Figure 3: Wilson action data structure on D_4 . (a) An elementary triangular plaquette $(x, x + n_1, x + n_2)$ with link variables U_1, U_2, U_3 on its three oriented edges; the plaquette holonomy is $U_{\Delta} = U_1 U_2 U_3$. (b) Counts per D_4 unit cell: one site, 12 edges (link variables), and 32 triangular plaquettes (holonomies).

7 Plaquette Stiffness and the Continuum Limit

7.1 Plaquette stiffness tensor

For a triangular plaquette $\Delta = (x, x+n_1, x+n_2)$, the antisymmetric *bivector* encoding its orientation in the lattice is

$$b_{\mu\nu}^{\Delta} = (n_1)_{\mu}(n_2)_{\nu} - (n_1)_{\nu}(n_2)_{\mu}. \quad (27)$$

The *plaquette stiffness tensor* is the sum of bivector products over all triangles incident to a fixed site:

$$T_{\mu\nu\rho\sigma} \equiv \sum_{\Delta \ni x} b_{\mu\nu}^{\Delta} b_{\rho\sigma}^{\Delta}. \quad (28)$$

This tensor controls the leading-order continuum limit of the Wilson action — it decides whether the recovered Yang–Mills action is isotropic, and what coefficient relates β to g^2 .

Theorem 4 (Plaquette stiffness on D_4). *For the 96 triangular plaquettes incident to any site of D_4 ,*

$$T_{\mu\nu\rho\sigma} = c(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}), \quad c = 48, \quad (29)$$

exactly. The proportionality is component-wise, not on-average-only: $T_{\mu\nu\rho\sigma}$ vanishes on every tensor structure other than $(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$.

Proof. Direct enumeration. The 96 triangles incident to the origin are indexed by ordered pairs (n_1, n_2) of D_4 nearest neighbors with $n_2 - n_1 \in \mathcal{N}(D_4)$ (the third edge of the triangle). For each pair we compute $b_{\mu\nu}^{\Delta}$ from Eq. (27) and accumulate $b \otimes b$ into $T_{\mu\nu\rho\sigma}$. Numerically, $T_{0101} = 48$ and every component of T matches $48(\delta\delta - \delta\delta)$ to machine precision; the Frobenius difference $\|T - c(\delta\delta - \delta\delta)\|$ is zero.

Analytically, the result follows from F_4 invariance of the sum over triangles at a vertex. Any rank-(2,2) tensor antisymmetric in the first pair and the second pair, invariant under $F_4 \supset SO(4)$, has to be proportional to the unique $SO(4)$ -invariant rank-(2,2) tensor of that symmetry type, $(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$. $\square$

Per unit cell (1/3 of triangles at any origin site, since each triangle touches 3 vertices), $T_{\mu\nu\rho\sigma}^{\text{cell}} = 16(\delta\delta - \delta\delta)$. Per unit 4-volume (cell volume = 2), the density is $T_{\mu\nu\rho\sigma}^{\text{density}} = 8(\delta\delta - \delta\delta)$.

7.2 Continuum limit

Expanding $U_e \approx \exp(igaA_\mu(n)_\mu)$ for an edge in direction n and evaluating the plaquette holonomy to second order in ga ,

$$\text{Re Tr } U_\Delta \approx N - \frac{1}{2}g^2 \text{Tr}(F_{\mu\nu}F_{\rho\sigma}) A_{\mu\nu}^\Delta A_{\rho\sigma}^\Delta + O(g^4, a^6), \quad (30)$$

where $A_{\mu\nu}^\Delta = \frac{1}{2}b_{\mu\nu}^\Delta$ is the signed area bivector. Substituting into the Wilson action Eq. (26) gives

$$\begin{aligned} S_G &= \beta \sum_{\Delta} \frac{1}{2N} g^2 \text{Tr}(F_{\mu\nu}F_{\rho\sigma}) A_{\mu\nu}^\Delta A_{\rho\sigma}^\Delta + O(g^4) \\ &= \frac{\beta g^2}{8N} \text{Tr}(F_{\mu\nu}F_{\rho\sigma}) T_{\mu\nu\rho\sigma}^{\text{total}}, \end{aligned} \quad (31)$$

with $T_{\mu\nu\rho\sigma}^{\text{total}}$ the stiffness summed over all plaquettes in the lattice. Using the per-volume density and the exact isotropy of Eq. (29),

$$S_G \xrightarrow{a \rightarrow 0} \frac{1}{4g_{\text{eff}}^2} \int d^4x \text{Tr}(F_{\mu\nu}F^{\mu\nu}) \quad (32)$$

with $g_{\text{eff}}^2 = \alpha/\beta$ for some numerical α that depends on the conventions used in tracking bivector normalizations.

What the isotropy guarantees. $T_{\mu\nu\rho\sigma} \propto (\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ is the only tensor of its symmetry type compatible with $SO(4)$. The leading-order continuum action is therefore Eq. (32) *without* added anisotropy terms like $\sum_\mu \text{Tr}(F_{\mu\nu}^2)$, which would break $SO(4)$ to a cubic subgroup the way the standard hypercubic action does at $O(a^2)$. On D_4 , the lattice corrections at $O(a^2)$ are absent at the level of the gauge action; the first deviations from continuum Yang–Mills enter at $O(a^4)$ through higher-derivative terms invariant under the residual lattice symmetry.

What the isotropy does not fix. The dimensionless prefactor α that relates g_{eff}^2 to β depends on how the bivector normalization, area normalization, and plaquette density are tracked through the expansion. We do not pin α to a specific number here. Numerical perturbative matching at one loop — comparing D_4 lattice perturbation theory to continuum $\overline{\text{MS}}$ at a chosen scale — is the standard way to fix this constant [6] and is the natural next step toward quantitative lattice work on D_4 .

Part IV

Wilson Fermions and Quarks as Defects

8 Wilson Fermions on D_4

8.1 Action and naive Dirac operator

The naive Dirac action of Section 4 couples to gauge links as

$$S_F^{\text{naive}} = \frac{1}{2a} \sum_x \sum_{n \in \mathcal{N}(D_4)} \bar{\psi}(x) \gamma_\mu \hat{n}_\mu U_{x,n} \psi(x + an) + \text{h.c.}, \quad (33)$$

where $\hat{n} = n/|n|$ is the unit vector and $U_{x,n}$ is the link variable connecting x to $x + an$. The momentum-space kinetic field is given by Theorem 2, with zero modes (doublers) classified by Theorem 3.

8.2 Wilson term

The standard Wilson term [4] on D_4 adds the lattice-Laplacian-like operator

$$W\psi(x) = \frac{r}{a} \sum_{n \in \mathcal{N}(D_4)} [\psi(x) - U_{x,n} \psi(x + an)] + \text{h.c.}, \quad (34)$$

giving the momentum-space Wilson mass

$$m_W(k) = \frac{2r}{a} \sum_{n \in \mathcal{N}(D_4)} (1 - \cos(k \cdot n)). \quad (35)$$

The Wilson parameter r is dimensionless; we work with the standard choice $r = 1$ throughout.

8.3 Wilson masses at every zero-mode class

Evaluating Eq. (35) at one representative momentum per class:

Proposition 4 (Wilson spectrum). *The Wilson mass values at the zero-mode classes of D_4 (with $r = 1$, $a = 1$) are*

Class	m_W (units of $1/a$)	Identification
$\Gamma = (0, 0, 0, 0)$	0	<i>physical fermion</i>
$X = (\pi, 0, 0, 0)$ and perms	48	<i>Case-1 doubler, $n_\pi = 1$</i>
$M = (\pi, \pi, 0, 0)$ and perms	64	<i>Case-1 doubler, $n_\pi = 2$</i>
$R = (\pi, \pi, \pi, 0)$ and perms	48	<i>Case-1 doubler, $n_\pi = 3$</i>
$\text{all-}\pi = (\pi, \pi, \pi, \pi)$	0	<i>identified with Γ, see below</i>
L_2	56	<i>Case-2 doubler, $n_C = 2$</i>
L_3	54	<i>Case-2 doubler, $n_C = 3$</i>
L_4	48	<i>Case-2 doubler, $n_C = 4$</i>

All non- Γ doublers have $m_W \geq 48/a$, and the physical fermion at Γ remains massless.

The full Wilson spectrum is shown in Figure 4.

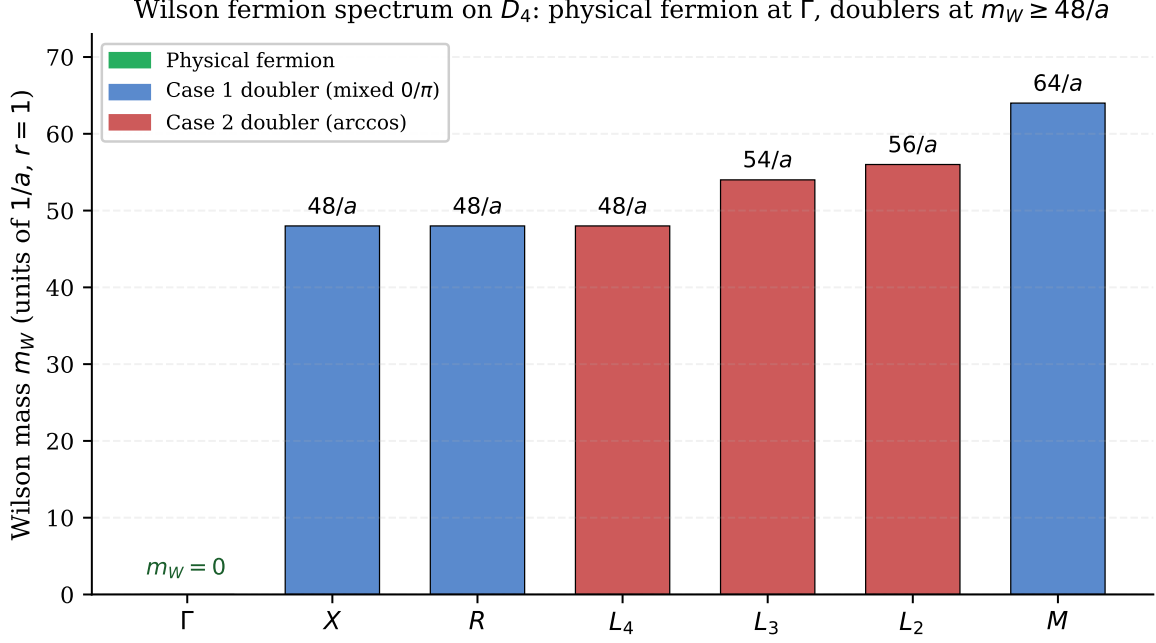


Figure 4: Wilson fermion mass spectrum on D_4 at $r = 1$. The physical fermion at Γ remains massless; all seven doubler classes acquire Wilson mass $m_W \geq 48/a$ and decouple in the continuum limit. Case-1 doublers (mixed $0/\pi$ momenta) and Case-2 doublers (arccos-determined momenta) span comparable mass ranges, with $X = R = L_4 = 48/a$ as the lightest doubler tier.

8.4 The all- π identification

The Wilson term vanishes at $k = (\pi, \pi, \pi, \pi)$ as well as at $\Gamma = (0, 0, 0, 0)$, which would naively give two physical massless fermions. But the reciprocal lattice of D_4 is

$$D_4^* = D_4 \cup (D_4 + \omega_4) = \mathbb{Z}^4 \cup (\mathbb{Z}^4 + (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})), \quad (36)$$

with fundamental weight $\omega_4 = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ generating the half-integer coset. The reciprocal vector $g = 2\pi\omega_4 = \pi(1, 1, 1, 1)$ therefore lies in $2\pi D_4^*$, and the all- π point is BZ-equivalent to Γ :

$$(\pi, \pi, \pi, \pi) - \pi(1, 1, 1, 1) = (0, 0, 0, 0). \quad (37)$$

The two zeros of the Wilson mass correspond to a single momentum in the first Brillouin zone, leaving one physical massless fermion. The same BZ-equivalence identification arises in the 3D FCC case via the corresponding half-integer reciprocal vector.

8.5 Doubler removal: status

The Wilson term as written lifts every doubler to a positive mass, and the all- π Case-1 zero is identified with Γ . The result, at the level of the explicit Wilson masses computed in Eq. (35), is exactly one physical fermion.

A complete analysis of doubler removal also requires the BZ equivalences within each Case-2 class. The 48 L_2 , 64 L_3 , and 16 L_4 raw zeros include pairs related by reciprocal-lattice translations — notably the $\pi(1, 1, 1, 1)$ shift, which permutes signs of $\pm\pi/2$ entries and shifts arccos angles. Enumerating BZ-inequivalent doubler points within each class, and checking that Nielsen–Ninomiya

holds on the reduced set, is a technical calculation that we leave to numerical lattice work. The raw chirality sum of Eq. (24) establishes the topological consistency of the spectrum; the equivalence-class accounting is the more delicate finishing step.

9 Quarks as Trapped Tetrahedral Defects

9.1 The two viewpoints

The lattice gauge theory constructed above admits two views of what a quark is. In the standard *bulk-field viewpoint*, $\psi(x)$ is a spinor at every D_4 site, carrying color, flavor, and spin indices, gauge-coupled through the link variables U_e . That is the conventional lattice QCD setup adapted to D_4 , and the Wilson fermion analysis of Section 8 is its concrete realization.

In the *defect viewpoint* of the matter paper [12], a physical quark is a localized lattice defect — a trapped extra node at the centroid of a tetrahedral void in a spatial FCC slice $x_4 = t$. The defect bonds to four FCC sites by the edges of the surrounding tetrahedron, and the matter paper builds color, charge, and confinement on that localized object. The point of this section is not to re-derive the matter paper but to check that its defects are compatible with the D_4 Wilson gauge action of Sections 6–8.

9.2 Compatibility with the Wilson action

The slicing theorem (Theorem 1) makes the compatibility almost immediate: any spatial defect configuration on the 3D FCC lattice embeds unchanged into the $x_4 = t$ slice of D_4 . A defect at spatial position $\vec{x}$ in slice t couples to two kinds of gauge link from the D_4 Wilson action:

- *In-slice links*: the spatial valence-bond links from the defect site to its FCC neighbors. These are the 12 spatial neighbors of Proposition 2.
- *Cross-slice links*: the 12 temporal neighbors connecting $(\vec{x}, t)$ to $(\vec{x} + \delta, t \pm 1)$ for $\delta \in \mathcal{N}_{\text{spatial}}/2$ (the diagonal time bonds).

The in-slice links carry the spatial gauge structure — the same $SU(3)$ link variables that act on the defect’s color state in the matter paper [12]. The cross-slice links propagate that color state from one time slice to the next.

Two structural conditions are needed for the viewpoints to agree:

1. The color Hilbert space of a single quark transforms under $SU(3)$ as a fundamental (**3**). This is the universal property of quarks across every formulation of QCD; it holds by construction in the matter paper [12].
2. The gauge dynamics on the defect’s valence bonds is the dynamics generated by the Wilson action Eq. (26) on those same bonds. This is automatic — the valence bonds are bonds of the underlying D_4 lattice, and the Wilson action assigns $SU(3)$ link variables to every such bond.

Both conditions hold. The first says that the local color rotations of the defect generate $\mathfrak{su}(3)$ in the **3**; the second says that those rotations are propagated across the lattice by the D_4 Wilson field. The two viewpoints describe the same $SU(3)$ gauge theory at different operational levels: the matter paper’s defect framework says what a quark *is* as a localized lattice configuration; the D_4 Wilson action says how the gauge field acts on it.

Appendix A closes the second half of the picture. It rebuilds $\mathfrak{su}(3)$ on the defect’s color Hilbert space from defect geometry alone — three colors from the skew-pair partition of K_4 , the Weyl group S_3 from the quotient S_4/V_4 under anchor selection, the rank-2 maximal torus from three Bell-pair $U(1)$ phases modulo the diagonal $U(1)$ that gauges to electromagnetism — and shows that the eight resulting generators close on $\mathfrak{su}(3)$ with the A_2 root system in the fundamental representation **3**. The three-color content itself — $\dim \mathcal{H}_C = 3$ from the three skew-edge pairs of the bounding tetrahedron — is the matter paper’s result [12, Section 5.1]; the algebraic completion is Appendix A.

Inheritance of geometric confinement. Because the two viewpoints describe the same gauge theory on the same lattice, the D_4 construction inherits the confinement structure of the matter paper [12, Section 6]. There, the linear σr piece of the Cornell potential follows from a static lattice obstruction: the metric wall at the in-plane circumradius $L/\sqrt{3}$ of a triangular face of the cuboctahedron stops a trapped tetrahedral defect from being extracted without breaking $O(r)$ bond connections. String breaking emerges when the bond-length tolerance is exceeded and the lattice nucleates a new vertex pair — a quark-antiquark pair. Electromagnetic invisibility of fractional charges follows from the non-bipartite topology of the 8 triangular faces of the cuboctahedron, which cannot support the alternating $\{+, -\}$ dipole modes that mediate photon exchange. None of these need a dynamical gauge field; they are static-topological properties of the FCC slice, and the D_4 Wilson action propagates them in time without modification. So the D_4 formulation is not merely a lattice discretization of $SU(3)$ Yang–Mills whose continuum limit happens to confine. It is a lattice in which confinement has a geometric origin at the lattice scale, with the dynamical Wilson-loop area law emerging in the long-wavelength limit under renormalization-group flow [12, Section 6.2].

9.3 Hadrons as bound defect configurations

Hadrons in the defect framework are bound configurations of defects, the binding mediated by the gauge field. The proton is three trapped tetrahedral defects bound into a color singlet. Mesons are defect-antidefect pairs. Computing hadron masses on D_4 — the proton-to-pion ratio m_p/m_π , for example — requires dynamical lattice simulations and is not undertaken here.

10 Discussion and Future Work

10.1 What this paper establishes

The structural results here are rigorous and reproducible. Proved or directly computed: the slicing structure (Theorem 1 and Proposition 2), the factorization theorem (Theorem 2), the zero-mode classification with Nielsen–Ninomiya verification (Theorem 3 and Eq. (24)), the Wilson action data (Proposition 3), the plaquette stiffness theorem (Theorem 4), and the Wilson fermion spectrum (Proposition 4). The geometric reconstruction of $\mathfrak{su}(3)$ from the defect (Appendix A) is established by explicit seven-step construction: skew-pair count, the $S_4/V_4 = S_3$ Weyl group from anchor selection (Proposition 5), the rank-2 maximal torus from Bell-pair phases modulo $U(1)_{\text{diag}}$, the Cartan generators, the root generators, the A_2 root system, and the structure constants. The verification script [16] reproduces every numerical claim, including the explicit check that $\ker(\pi : S_4 \rightarrow S_3) = V_4$ by enumeration over all 24 permutations.

10.2 What is deferred to numerical work

Four technical questions are left for numerical lattice simulation.

The exact value of α . The dimensionless prefactor α in $g_{\text{eff}}^2 = \alpha/\beta$ depends on conventions for bivector and plaquette area normalizations. Fixing it requires perturbative matching to continuum $\overline{\text{MS}}$ at one loop, a standard [6] but explicit lattice perturbation theory calculation beyond the scope of this structural paper.

BZ-equivalence-class chirality verification. The raw chirality sum $0 - 48 + 64 - 16 = 0$ of Eq. (24) confirms Nielsen–Ninomiya at the level of all 144 zero momenta. Reducing modulo BZ equivalences (notably the $\pi(1, 1, 1, 1)$ shift) gives a smaller set of inequivalent zeros, whose chirality sum should also vanish; we did not complete that reduction here.

Continuum-limit corrections. The leading-order continuum action is isotropic. The explicit form of the $O(a^2)$ and $O(a^4)$ corrections — typically small on D_4 thanks to the high coordination number — is what would matter for high-precision matching to continuum $SU(3)$ Yang–Mills.

Hadronic spectrum. The eventual validation of D_4 lattice QCD is the calculation of hadron masses and decay constants in dynamical simulations and the comparison with experiment. That requires implementing the Wilson action of Section 8 with dynamical quarks on a D_4 Monte Carlo lattice, beyond the present analytical work.

10.3 Comparison with hypercubic lattice gauge theory

The standard hypercubic $\mathbb{Z}^4$ lattice has been the workhorse of lattice QCD for five decades, as a numerical discretization of continuum spacetime. The comparison this paper wants to make is different: not which 4D lattice gives the most efficient discretization, but which 4D lattice could be the physical substrate of spacetime. Table 2 summarizes the comparison.

Structural fit as physical substrate. Six properties separate the two candidates.

The slicing theorem (Theorem 1) gives $D_4 \cap \{x_4 = 0\} = \text{FCC}$ — every constant-time slice of D_4 is FCC. Hypercubic $\mathbb{Z}^4$ has $\mathbb{Z}^3$ slices, not FCC. Under SSM, the spatial substrate is FCC at every time, so hypercubic is not a candidate at all.

D_4 is the densest 4D packing [2] with maximum kissing number $K = 24$. $\mathbb{Z}^4$ has $K = 8$. For a physical substrate, maximum packing density and maximum coordination correspond to maximum rigidity and finest angular resolution at short distances.

The D_4 structure tensor with unit-length normalization is $\hat{S}_{\mu\nu} = 6\delta_{\mu\nu}$ (Proposition 1), three times the hypercubic value $\hat{S}_{\mu\nu} = 2\delta_{\mu\nu}$. Both are isotropic, but D_4 carries more weight in each direction — the discrete analog of having more terms in a 4D Gauss quadrature rule.

Both lattices are self-dual, but their symmetry groups are not the same size. The Weyl group F_4 of order 1152 is three times the hypercubic point group B_4 of order 384, leaving more residual lattice symmetry on observables and operators.

Gauge action structure. The 24-cell coordination polytope of D_4 has only triangular faces, so the Wilson action is single-coupling with 32 triangular plaquettes per unit cell. Hypercubic has only square plaquettes, 6 per unit cell.

Property	D_4 (this paper)	Hypercubic $\mathbb{Z}^4$
<i>Structural fit as physical substrate</i>		
Contains 3D FCC as a slice?	Yes (Thm. 1)	No
Densest 4D packing?	Yes	No
Kissing number K	24 (maximum in 4D)	8
Structure tensor isotropy	$\hat{S}_{\mu\nu} = 6\delta_{\mu\nu}$ (Prop. 1)	$\hat{S}_{\mu\nu} = 2\delta_{\mu\nu}$
Reciprocal lattice	Self-dual, $D_4^* = D_4$ up to scale	Self-dual, $\mathbb{Z}^4$
Point/Weyl group order	$ F_4 = 1152$	$ B_4 = 384$
<i>Gauge action structure</i>		
Plaquette types	Triangular only	Square only
Gauge couplings	Single β	Single β
Plaquettes per unit cell	32 triangles	6 squares
Plaquette stiffness tensor	Exact $(\delta\delta - \delta\delta)$ (Thm. 4)	Isotropic + $O(a^2)$ cubic-anisotropic piece
Effective $SO(4)$ symmetry at long wavelengths	Exact at every order in a	Approximate; cubic anisotropy at $O(a^2)$
<i>Fermion spectrum</i>		
Fermion hops per site	24	8
Raw doublers in BZ	144 in 4 classes (Thm. 3)	16 in 1 class
Wilson doubler lifting	All doublers $\geq 48/a$ (Prop. 4)	Doublers at fixed mass $\sim 8/a$
<i>Connection to defect picture</i>		
FCC defects realizable in slices?	Yes (slicing theorem)	No (no FCC slice)
$\mathfrak{su}(3)$ generators on $\mathcal{H}_C = \mathbb{C}^3$	Self-contained on FCC slices (App. A)	Not applicable

Table 2: Comparison of D_4 and hypercubic $\mathbb{Z}^4$ as candidate physical structures of 4D spacetime. Under the SSM framework, the structural-fit criteria (first block) and the consistency with the spatial defect picture (last block) make D_4 the unique consistent choice. The intermediate blocks describe properties of the resulting gauge theory.

The plaquette stiffness theorem (Theorem 4) gives an *exact* $(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ tensor structure on D_4 . The effective long-wavelength action at scales $L \gg a$ is Yang–Mills with exact $SO(4)$ Euclidean symmetry — equivalently, exact Lorentz invariance under Wick rotation — at every order in a . On hypercubic, the analogous tensor has an isotropic part plus a cubic-anisotropic piece at order a^2 , breaking $SO(4)$ to the cubic group at sub-Planckian scales [5].

Under SSM, a is a fixed physical scale of order the Planck length, and the distinction does not go away by taking $a \rightarrow 0$ — the spacing is what it is. On D_4 , the effective theory at long wavelengths is exactly Lorentz-invariant. On hypercubic, residual cubic anisotropy is present at every sub-Planckian scale, suppressed only by powers of $(\text{energy}/E_{\text{Planck}})$. The exact-isotropy result on D_4 therefore has a physical meaning under SSM that it does not have under the standard interpretation, where the anisotropy disappears by construction in the continuum limit.

Fermion spectrum. Wilson fermions on D_4 hop to 24 nearest neighbors, giving a richer spectrum than the 8-neighbor hypercubic hop. The naive Dirac operator has 144 raw doublers in 4 topological classes (Theorem 3), versus 16 doublers in a single class on hypercubic. The Wilson term lifts every D_4 doubler to mass $m_W \geq 48/a$ (Proposition 4); the physical fermion at Γ remains massless.

The four-class doubler structure on D_4 is more intricate than the uniform hypercubic case and warrants further study, particularly for the BZ-equivalence reduction of Section 8.

Connection to the defect picture. Physical matter in SSM is built from FCC defects — specifically, quarks are trapped tetrahedral defects in FCC slices [12]. The construction requires the spatial substrate to be FCC. On D_4 , every constant-time slice is FCC, so the defect picture extends without modification. On hypercubic, the spatial slices are $\mathbb{Z}^3$, and the tetrahedral defect construction does not apply — $\mathbb{Z}^3$ has no tetrahedral voids of the type the matter paper [12] uses. Appendix A reconstructs the $\mathfrak{su}(3)$ algebra acting on the defect’s three-dimensional color Hilbert space from defect geometry. The construction is meaningful on D_4 via the FCC slicing; not on $\mathbb{Z}^4$.

Net assessment. Under SSM, D_4 is the unique 4D lattice consistent with the discrete-substrate program: it preserves the physical spatial FCC structure at every time, it has the maximum 4D coordination, and it admits the tetrahedral defect picture that the matter paper [12] establishes for SM matter content. Hypercubic is geometrically inconsistent with the framework — not because it is computationally inferior but because it does not contain FCC as a slice and does not support the defect picture.

For readers coming from the standard lattice QCD viewpoint, the same construction is an alternative discretization with exact $SO(4)$ isotropy at every order, of independent mathematical interest. The analytical results — slicing theorem, exact Dirac-operator factorization, plaquette stiffness theorem, connection to the defect picture — characterize the continuum behavior of the D_4 formulation without numerical simulation. The mathematics of Sections 3–8 and Appendix A is identical under either reading; only the physical meaning of a and of the continuum limit differs.

10.4 Distinguishing predictions of the framework

Four features of the D_4 construction make falsifiable predictions that distinguish the SSM framework from continuum quantum field theory.

Gauge group forced, not chosen. In continuum gauge theory, the choice of $SU(N)$ for the strong sector is a free input — $SU(3)$ is selected by experiment, not derived. The construction here goes the other way. Appendix A reconstructs the full $\mathfrak{su}(3)$ Lie algebra of the strong sector from the geometry of a trapped tetrahedral defect: three colors from the skew-pair partition of K_4 via $\binom{4}{2}/2 = 3$, Weyl group $W(A_2) = S_3$ from the quotient S_4/V_4 under anchor selection (the same anchor selection that produces the up- and down-quark charges in the matter paper [12]), and the rank-2 maximal torus from three Bell-pair $U(1)$ phases modulo the diagonal $U(1)$ that the matter paper identifies with electromagnetism. The eight generators close on $\mathfrak{su}(3)$ with the A_2 root system. A defect with different coordination would give a different residual symmetry, a different number of bond phases, and therefore a different Lie algebra by the same construction. Within the SSM identification of the trapped tetrahedral defect with the standard quark, the gauge group of the strong sector is geometrically determined, not empirically assigned. (The dynamical content of QCD — g_s , its running, Λ_{QCD} , asymptotic freedom — belongs to the bond energetics of the lattice and is not part of this static algebraic construction; see Section A.10.)

Confinement has a geometric origin. In continuum $SU(3)$ Yang–Mills, color confinement is a non-perturbative dynamical phenomenon: it is observed numerically in lattice QCD as a Wilson-loop area law, but its derivation from first principles in the continuum is still an open Millennium Problem. Standard hypercubic lattice QCD inherits the same situation: confinement is a dynamical consequence of strong coupling, not a property of the lattice geometry. In the D_4 formulation the situation is different. The spatial substrate is FCC at every time slice (Theorem 1), and on FCC the matter paper [12, Section 6] derives the linear σr piece of the Cornell potential as a static topological obstruction at the metric wall $L/\sqrt{3}$ — with string breaking from lattice un-stitching, and electromagnetic invisibility of fractional charges from the non-bipartite topology of triangular faces. Section 9 establishes that the bulk-field and defect viewpoints describe the same gauge theory, so the D_4 construction inherits this geometric confinement at the lattice scale, with the continuum Wilson-loop area law emerging in the long-wavelength limit. The framework’s prediction: confinement is not an emergent dynamical accident of strong coupling. It is a built-in geometric feature of the substrate. That is a qualitatively different claim from the one made by continuum QCD or hypercubic lattice QCD.

Wilson doublers as physical Planck-scale fermions. In standard lattice QCD, fermion doublers are artifacts removed by taking $a \rightarrow 0$. Under SSM, a is a fixed physical scale of order the Planck length and the limit is not available — the Wilson doublers are physical fermionic states with explicit masses $m_W \in \{48, 54, 56, 64\}/a$ (Proposition 4). At sub-Planckian energies they are inaccessible: their effects on low-energy observables are suppressed by $(E/M_P)^4$, far below current experimental reach. They are part of the framework’s physical fermionic spectrum, not computational artifacts to be removed.

Null Lorentz violation at leading order. Many discrete-spacetime models — hypercubic lattice formulations, several Planck-scale discrete-substrate scenarios — predict Lorentz violation at order $(E/M_P)^2$, motivating searches for energy-dependent photon arrival times in ultra-high-energy astrophysical sources. The plaquette stiffness theorem (Theorem 4) gives exact $(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ tensor structure with no $O(a^2)$ cubic-anisotropic piece, so the leading-order continuum gauge action is exactly $SO(4)$ -symmetric. The framework predicts a null result for $O(a^2)$ Lorentz-violation searches in the gauge sector. Whether higher-order corrections ($O(a^4)$ and beyond) preserve exact isotropy is left to future analysis (Section 10).

11 Conclusion

We have presented the D_4 root lattice as physical four-dimensional spacetime in the SSM framework and built the $SU(3)$ Wilson lattice gauge theory it supports. The geometric results — kissing number $K = 24$ (the maximum in $\mathbb{R}^4$), structure-tensor isotropy $\hat{S}_{\mu\nu} = 6\delta_{\mu\nu}$ (Proposition 1), the slicing $D_4 \cap \{x_4 = 0\} = \text{FCC}$ with $24 = 12 + 12$ neighbor decomposition, the Dirac factorization $f_\mu = (4/a)\sin(k_\mu)\sum_{\nu \neq \mu} \cos(k_\nu)$, and the four-class zero-mode structure with $\sum \chi = 0$ — are intrinsic to D_4 and recover the corresponding FCC structures on each time slice. The Wilson gauge action takes a single-coupling form on the 32 triangular plaquettes per unit cell, with plaquette stiffness tensor exactly proportional to $(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho})$ and coefficient $c = 48$. The effective long-wavelength action at scales $L \gg a$ is $SO(4)$ -symmetric Yang–Mills with no anisotropy correction at any sub-Planckian scale — a property that under SSM is a physical statement, not a continuum-limit artifact. Wilson fermions on D_4 have one massless physical mode at Γ (identified with the all- π zero via $2\pi D_4^*$ equivalence) and lift all doublers to $m_W \in \{48, 54, 56, 64\}/a$. Appendix A rebuilds

the full $\mathfrak{su}(3)$ Lie algebra acting on the defect’s color Hilbert space from the geometry of the trapped tetrahedral defect alone — three colors from the skew-pair partition of K_4 , Weyl group from anchor selection, rank-2 maximal torus from Bell-pair phases modulo the diagonal electromagnetic $U(1)$ — completing the connection between the local defect color content of the matter paper [12] and the lattice-wide $SU(3)$ Wilson gauge action.

What this gives is the gauge-theoretic backbone of the SSM framework at the 4D level. What remains for further work inside the program: the precise value of the prefactor α in $g_{\text{eff}}^2 = \alpha/\beta$, the BZ-reduced chirality enumeration, and the hadronic spectrum of bound defect configurations.

Acknowledgments

This work builds on the FCC lattice studies of [12, 11] and the broader SSMTheory program.

Data availability

A Python script that verifies all numerical claims of this paper — the isotropy of the structure tensor, the slicing decomposition, the Dirac factorization theorem, the zero-mode classification with chirality assignments, the Nielsen–Ninomiya sum, the per-unit-cell action data, the plaquette stiffness coefficient $c = 48$, the exact $(\delta\delta - \delta\delta)$ proportionality of $T_{\mu\nu\rho\sigma}$, the Wilson masses at every zero-mode class, and the $\mathfrak{su}(3)$ algebra of Appendix A (structure constants, Jacobi identity, Casimir) — is publicly available at https://github.com/raghu91302/ssmtheory/blob/main/verify_d4.py. No other data were generated or analyzed in this study.

A The $\mathfrak{su}(3)$ Algebra of a Trapped Tetrahedral Defect

This appendix builds $\mathfrak{su}(3)$ on the color Hilbert space of a trapped tetrahedral defect from defect geometry — not by importing $\mathfrak{su}(3)$ from textbook Lie theory and letting it act on $\mathbb{C}^3$, but by recovering each piece of the algebra from a separate feature of the defect: three colors from the skew-pair partition of K_4 , the Weyl group S_3 from the quotient S_4/V_4 under anchor selection, the rank-2 maximal torus from three Bell-pair $U(1)$ phases modulo the diagonal $U(1)$ that gauges to electromagnetism. The eight resulting generators close on $\mathfrak{su}(3)$ with the Cartan–Weyl structure constants of the A_2 root system, in the fundamental representation **3**. The construction is static and algebraic. The dynamical content of QCD — coupling g_s , running, Λ_{QCD} , asymptotic freedom — belongs to the bond energetics of the lattice and is not part of this appendix. The full treatment, including the complete 28-commutator closure table verified by direct matrix computation, is in the companion preprint [14].

A.1 The defect color Hilbert space from skew pairs

A tetrahedral void in the FCC lattice is bounded by four FCC sites at the vertices of a regular tetrahedron. A *trapped extra node* placed at the centroid bonds to all four bounding sites; we label these bonds b_1, b_2, b_3, b_4 . Each carries a $U(1)$ phase degree of freedom inherited from the Bell-pair structure on FCC nearest-neighbor bonds [12, 11], an entangled phase between the centroid node and its FCC endpoint. The four phases will feed both the color Hilbert space (here) and the maximal torus (Section A.3).

The matter paper [12, Section 5.1] gives the combinatorial origin of three colors: the complete graph K_4 on the four bond labels has exactly three skew-edge pairs (equivalently, three perfect

matchings),

$$P_1 = \{\{b_1, b_2\}, \{b_3, b_4\}\}, \quad P_2 = \{\{b_1, b_3\}, \{b_2, b_4\}\}, \quad P_3 = \{\{b_1, b_4\}, \{b_2, b_3\}\}, \quad (38)$$

and that count is rigid: $\binom{4}{2}/2 = 3$, with no fourth pairing possible. (Equivalently, K_{2n} has $(2n-1)!!$ perfect matchings, giving $3!! = 3$ for K_4 .) We associate to each skew pairing P_i a quantum state $|i\rangle$ encoding the phase-coherent superposition of its two paired Bell-pair phases. The three states span a 3-dimensional complex Hilbert space:

$$\mathcal{H}_C = \text{span}_{\mathbb{C}}\{|1\rangle, |2\rangle, |3\rangle\} \cong \mathbb{C}^3, \quad \langle i|j\rangle = \delta_{ij}. \quad (39)$$

The dimension is fixed by tetrahedral combinatorics, not chosen: $\dim \mathcal{H}_C = 3$ because K_4 has exactly three perfect matchings. A defect with different coordination would give a different dimension. The tetrahedral void is what forces three colors.

A.2 The Weyl group from anchor selection

The full permutation group of the four bond labels $\{b_1, b_2, b_3, b_4\}$ is the symmetric group S_4 of order 24. Each element $\sigma \in S_4$ acts on edges of K_4 by permuting endpoints, $\sigma(\{b_i, b_j\}) = \{\sigma(b_i), \sigma(b_j)\}$, and consequently on skew pairs by acting on each edge of the pair. Since vertex-disjointness is permutation-invariant, $\sigma(P_i)$ is again a skew pair, giving a homomorphism

$$\pi : S_4 \longrightarrow \text{Sym}(\{P_1, P_2, P_3\}) \cong S_3. \quad (40)$$

Proposition 5 (Kernel of π). $\ker(\pi) = V_4 = \{e, (b_1 b_2)(b_3 b_4), (b_1 b_3)(b_2 b_4), (b_1 b_4)(b_2 b_3)\}$, the Klein four-group.

Proof. Direct enumeration. For each non-identity element of V_4 , the corresponding double transposition exchanges the two edges within each skew pair P_i , leaving P_i fixed as an unordered pair-of-pairs. For any $\sigma \in S_4 \setminus V_4$, σ permutes the three skew pairs non-trivially: e.g., the transposition $(b_1 b_2)$ sends $P_2 = \{\{b_1, b_3\}, \{b_2, b_4\}\}$ to $\{\{b_2, b_3\}, \{b_1, b_4\}\} = P_3$. $\square$

By the first isomorphism theorem,

$$S_4/V_4 \cong \text{Im}(\pi) = S_3, \quad (41)$$

where the image is all of S_3 because every transposition of skew pairs is realized by some element of S_4 .

Anchor selection. The matter paper [12, Section 4.1] introduces *anchor selection*: one of the four valence bonds is distinguished as the bulk-coupling channel through which charge, momentum, and color flux are exchanged with the surrounding $K = 12$ FCC lattice. The same anchor selection is what produces the up- and down-quark charge assignments in the matter paper. Without loss of generality the anchor is b_1 (the choice of which bond is the anchor is dynamical; a different choice gives the spatially-inverted defect). The stabilizer of b_1 in S_4 is the symmetric group on $\{b_2, b_3, b_4\}$, isomorphic to S_3 . Composing the inclusion $S_3 \hookrightarrow S_4$ with π gives an isomorphism

$$S_3 \xrightarrow{\sim} S_4/V_4, \quad (42)$$

because no non-trivial permutation of $\{b_2, b_3, b_4\}$ lies in V_4 . The residual symmetry of the defect after anchor selection is therefore precisely $S_4/V_4 \cong S_3$.

Identification with the Weyl group of $\mathfrak{su}(3)$. The Weyl group of $\mathfrak{su}(3)$ — the symmetry group of its A_2 root system, generated by reflections through the three pairs of opposite roots — is

$$W(A_2) = S_3. \quad (43)$$

The defect's residual S_3 symmetry from Eq. (42) is therefore the Weyl group of $\mathfrak{su}(3)$ acting on the three color states $\{|1\rangle, |2\rangle, |3\rangle\}$ — the same group acting on the same three objects, not an analog or a finite approximation.

A.3 The maximal torus from Bell-pair phases

A Bell-pair entanglement state on two qubits has the general form

$$|\Phi(\phi)\rangle = \frac{1}{\sqrt{2}}(|00\rangle + e^{i\phi}|11\rangle), \quad (44)$$

parametrized by a single real phase $\phi \in [0, 2\pi)$. The matter paper's identification of FCC bonds with Bell-pair entanglement [12, Section 2.3] assigns one continuous phase $\phi \in U(1)$ to each bond.

After anchor selection, the three valence bonds b_2, b_3, b_4 each carry an independent phase $\phi_2, \phi_3, \phi_4 \in U(1)$, and the full phase space is the 3-torus

$$T^3 = U(1)_2 \times U(1)_3 \times U(1)_4. \quad (45)$$

The diagonal subgroup

$$U(1)_{\text{diag}} = \{(e^{i\theta}, e^{i\theta}, e^{i\theta}) : \theta \in [0, 2\pi)\} \subset T^3 \quad (46)$$

shifts all three valence-bond phases by the same amount. On a localized state $|\psi\rangle \in \mathcal{H}_C$, $U(1)_{\text{diag}}$ multiplies every basis state $|i\rangle$ by the same overall phase $e^{i\theta}$. The matter paper [12, Section 4] identifies this diagonal $U(1)$ with electromagnetism: the conserved charge associated with $U(1)_{\text{diag}}$ is the integer winding number $W \in \{0, 1, 2, 3\}$ that produces the baryon charges $Q = -1 + W$.

Quotient $U(1)_{\text{diag}}$ out of T^3 and what is left is the rank-2 torus

$$T^2 \equiv T^3 / U(1)_{\text{diag}} = \{(\phi_2, \phi_3, \phi_4) : \phi_2 + \phi_3 + \phi_4 \equiv 0 \pmod{2\pi}\}. \quad (47)$$

The dimension of T^2 matches the rank of $\mathfrak{su}(3)$. Any maximal abelian subalgebra of $\mathfrak{su}(3)$ consists of diagonal traceless Hermitian matrices on $\mathbb{C}^3$, which form a 2-dimensional space. The defect carries exactly the maximal torus of $\mathfrak{su}(3)$.

A.4 Cartan generators

A natural basis of infinitesimal generators for the geometric torus T^2 of Eq. (47), expressed as diagonal traceless Hermitian matrices on $\mathcal{H}_C$, is

$$H_1 = \text{diag}(1, -1, 0), \quad H_2 = \frac{1}{\sqrt{3}} \text{diag}(1, 1, -2). \quad (48)$$

Both are diagonal, so they commute. Both are traceless, so they sit in $\mathfrak{su}(3)$ rather than $\mathfrak{u}(3)$ — the $U(1)_{\text{diag}}$ direction is gone. The diagonal vectors $(1, -1, 0)$ and $\frac{1}{\sqrt{3}}(1, 1, -2)$ are linearly independent in $\mathbb{R}^3$, and together they span the 2-dimensional traceless diagonal subalgebra.

Geometric meaning. The infinitesimal action of H_1 on a color state $|j\rangle$ is $H_1|j\rangle = (H_1)_{jj}|j\rangle$: $|1\rangle$ picks up phase $+1$, $|2\rangle$ picks up phase -1 , and $|3\rangle$ picks up phase 0 . H_1 therefore measures the phase difference between the skew-pair states $|1\rangle$ and $|2\rangle$, leaving the third state $|3\rangle$ untouched.

Similarly, H_2 assigns phase $1/\sqrt{3}$ to $|1\rangle$ and $|2\rangle$ and phase $-2/\sqrt{3}$ to $|3\rangle$, measuring the phase imbalance between $|3\rangle$ on the one hand and the symmetric combination of $|1\rangle$ and $|2\rangle$ on the other. Each color state is itself a phase-coherent superposition of the underlying Bell-pair bond phases (Eq. (39)), so H_1 and H_2 correspond to specific linear combinations of those bond phases that are orthogonal (after the $U(1)_{\text{diag}}$ quotient of Section A.3) to the diagonal electromagnetic direction.

The factor of $1/\sqrt{3}$ in H_2 is the normalization that makes H_1, H_2 orthogonal under the Killing form $K(X, Y) = 2 \text{tr}(XY)$ on $\mathfrak{su}(3)$, giving $K(H_i, H_j) = 4\delta_{ij}$. Without it the algebra is unchanged, but the roots come out at unequal lengths and the A_2 angles get obscured.

Relation to the standard QCD-normalized basis. In the QCD-standard normalization $T^a = \lambda^a/2$ with the Gell-Mann matrices λ^a , the Cartan generators are

$$T^3 = \frac{1}{2}H_1, \quad T^8 = \frac{1}{2}H_2, \quad (49)$$

satisfying $\text{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. Sections A.5–A.8 use the QCD-standard basis $\{T^a\}_{a=1}^8$ to make contact with the standard $\mathfrak{su}(3)$ structure constants and the body of this paper. The geometric content (Eq. (48)) is the same in either normalization, up to a factor of 2.

A.5 Root generators from color transitions

A root vector of $\mathfrak{su}(3)$ raises or lowers between weight spaces (eigenspaces of the Cartan subalgebra). In the color basis $\{|1\rangle, |2\rangle, |3\rangle\}$, the off-diagonal matrix units $E_{ij} = |i\rangle\langle j|$ for $i \neq j$ implement transitions between distinct color states — E_{ij} takes the trapped node from color j to color i . In QCD terms, this transition is the local algebraic content of a gluon emission/absorption vertex on the defect. It is not the propagating gluon itself, which is a degree of freedom living on the FCC bulk between defects with a dispersion relation set by the bond energetics (and outside the scope of this construction).

There are $3 \times 2 = 6$ such off-diagonal generators $\{E_{12}, E_{13}, E_{23}, E_{21}, E_{31}, E_{32}\}$. With the 2 Cartan generators, that gives 8 in total, matching $\dim \mathfrak{su}(3) = 3^2 - 1 = 8$.

In the QCD-standard normalization, the six off-diagonal Gell-Mann/2 generators

$$T^1, T^2 \in \text{span}_{\mathbb{R}}\{E_{12}, E_{21}\}, \quad T^4, T^5 \in \text{span}_{\mathbb{R}}\{E_{13}, E_{31}\}, \quad T^6, T^7 \in \text{span}_{\mathbb{R}}\{E_{23}, E_{32}\}, \quad (50)$$

together with T^3, T^8 from Section A.4, are the explicit Hermitian basis

$$\begin{aligned} T^1 &= \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & T^2 &= \frac{1}{2} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & T^3 &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\ T^4 &= \frac{1}{2} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & T^5 &= \frac{1}{2} \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & T^6 &= \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ T^7 &= \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & T^8 &= \frac{1}{2\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \end{aligned} \quad (51)$$

The geometric interpretation: $T^{1,2}$ generate $|1\rangle \leftrightarrow |2\rangle$ exchanges (rotations and phase shifts in the $\{P_1, P_2\}$ subspace of skew pairs); $T^{4,5}$ generate $|1\rangle \leftrightarrow |3\rangle$ exchanges; $T^{6,7}$ generate $|2\rangle \leftrightarrow |3\rangle$ exchanges; T^3 and T^8 are the Cartan generators of Section A.4.

A.6 The A_2 root system

For each off-diagonal generator E_{ij} , the commutator with a Cartan generator $H \in \text{span}\{H_1, H_2\}$ has the form $[H, E_{ij}] = \alpha_{ij}(H)E_{ij}$ with eigenvalue $\alpha_{ij}(H)$ linear in H . Direct computation gives

$$\begin{aligned} [H_1, E_{12}] &= +2E_{12}, & [H_1, E_{13}] &= +1E_{13}, & [H_1, E_{23}] &= -1E_{23}, \\ [H_2, E_{12}] &= 0, & [H_2, E_{13}] &= +\sqrt{3}E_{13}, & [H_2, E_{23}] &= +\sqrt{3}E_{23}, \end{aligned} \quad (52)$$

together with the negatives for $E_{ji} = E_{ij}^\dagger$. The columns of Eq. (52) are the root coordinates in the (H_1, H_2) plane:

$$\alpha_{12} = (2, 0), \quad \alpha_{13} = (1, \sqrt{3}), \quad \alpha_{23} = (-1, \sqrt{3}), \quad (53)$$

together with $-\alpha_{12}, -\alpha_{13}, -\alpha_{23}$. All six roots have $|\alpha|^2 = 4$ (in the H_i basis), and the inner products $\alpha_{12} \cdot \alpha_{13} = 2$, $\alpha_{12} \cdot \alpha_{23} = -2$, $\alpha_{13} \cdot \alpha_{23} = 2$ give cosines $\pm 1/2$ — angles 60° and 120° between adjacent roots. That is the A_2 root system, the unique rank-2 simply-laced root system with all roots of equal length and angles in $\{60^\circ, 120^\circ, 180^\circ\}$.

By the classification of complex semisimple Lie algebras, the algebra with root system A_2 is $\mathfrak{su}(3)$ uniquely (up to isomorphism over $\mathbb{C}$, with real form $\mathfrak{su}(3)$ fixed by Hermiticity of the generators).

A.7 Structure constants and closure

In the QCD-standard normalization Eq. (51), the commutators of the eight generators close under linear combinations,

$$[T^a, T^b] = i f^{abc} T^c, \quad (54)$$

with totally antisymmetric structure constants f^{abc} . Direct computation (or evaluation of $f^{abc} = -2i \text{tr}(T^c[T^a, T^b])$ using $\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$) gives the non-zero structure constants

$$\begin{aligned} f^{123} &= 1, \\ f^{147} = f^{246} = f^{257} = f^{345} &= \frac{1}{2}, & f^{156} = f^{367} &= -\frac{1}{2}, \\ f^{458} = f^{678} &= \frac{\sqrt{3}}{2}. \end{aligned} \quad (55)$$

These are exactly the structure constants of $\mathfrak{su}(3)$ in the standard Gell-Mann basis. The Jacobi identity

$$[T^a, [T^b, T^c]] + [T^b, [T^c, T^a]] + [T^c, [T^a, T^b]] = 0 \quad (56)$$

holds for all $\binom{8}{3} = 56$ ordered triples by direct computation. All 28 independent commutators close on the 8-dimensional span of $\{T^a\}_{a=1}^8$; the full commutator table is in [14].

A.8 Quadratic Casimir and the fundamental representation

The action of the eight generators on $\mathcal{H}_C \cong \mathbb{C}^3$ has weights (eigenvalues of (H_1, H_2) on the basis vectors $|i\rangle$)

$$\mu_1 = (1, \frac{1}{\sqrt{3}}), \quad \mu_2 = (-1, \frac{1}{\sqrt{3}}), \quad \mu_3 = (0, -\frac{2}{\sqrt{3}}), \quad (57)$$

forming an equilateral triangle of side $|\mu_i - \mu_j|^2 = 4$ centered at the origin (since $\mu_1 + \mu_2 + \mu_3 = 0$) — the weight diagram of the fundamental representation **3** of $\mathfrak{su}(3)$.

The quadratic Casimir is

$$C_2 = \sum_{a=1}^8 T^a T^a = \frac{4}{3} \mathbb{1}, \quad (58)$$

with $\mathbb{I}$ the identity on $\mathcal{H}_C$. The eigenvalue $C_2 = 4/3$ is the standard value for $\mathfrak{su}(3)$ in the fundamental (**3**). The matter paper’s identification of the three skew pairs with QCD color charges therefore says exactly this: the trapped node, viewed as a localized state in $\mathcal{H}_C$, transforms in the fundamental representation of the geometrically constructed $\mathfrak{su}(3)$.

A.9 Why $\mathfrak{su}(3)$ and only $\mathfrak{su}(3)$

The construction above gives, with no free parameter and no external algebraic input:

1. Three colors, from the three skew-edge pairs of K_4 (Section A.1).
2. The Weyl group S_3 , from the quotient S_4/V_4 induced by anchor selection (Section A.2).
3. The rank-2 maximal torus T^2 , from the three valence-bond $U(1)$ phases modulo the diagonal $U(1)$ that gauges to electromagnetism (Section A.3).
4. Two Cartan and six root generators, totaling 8 generators (Sections A.4–A.5).
5. The A_2 root system at the correct lengths and angles (Section A.6).
6. The Cartan–Weyl structure constants of $\mathfrak{su}(3)$ (Section A.7).
7. The fundamental representation **3** on the three colors (Section A.8).

These seven pieces fix *both* the representation space ($\mathbb{C}^3$, with the geometric basis $\{|i\rangle\}_{i=1}^3$ from skew pairs) *and* the algebra acting on it ($\mathfrak{su}(3)$, with the geometric Weyl group from anchor selection and the geometric maximal torus from Bell-pair phases). The algebra is not imported from textbook Lie theory; standard Lie theory enters only to identify what was constructed — the unique compact simple Lie algebra of rank 2 with the A_2 root system is $\mathfrak{su}(3)$ [15].

A defect with different coordination yields a different algebra by the same procedure. The octahedral defect bounded by six FCC sites, for instance, would have $\binom{6}{2}/2 \cdot$ (combinatorial factor) skew structure, a different residual symmetry after anchor selection, and a different number of bond phases — a different Lie algebra. This matches the matter paper’s identification of octahedral defects with a distinct sector [12].

A.10 What this construction does not deliver

The construction is static and algebraic. It fixes the gauge group of the strong sector and the representation in which quarks transform, but it does not deliver the dynamical content of QCD:

- The coupling g_s and its running with energy scale.
- Asymptotic freedom and Λ_{QCD} .
- Confinement as a Wilson-loop area law (the matter paper [12, Section 6] derives a static topological version of confinement at the metric wall $L/\sqrt{3}$; the dynamical area law is a separate matter, emerging in the continuum limit when bond fluctuations are taken into account).
- Higher quark generations. The construction is a first-shell defect classification; second-shell extensions and heavier flavors require the broader spectrum analysis of [13].

These dynamical features live in the bond energetics of the FCC lattice. The Wilson gauge action of Section 6 is what propagates the locally defined $\mathfrak{su}(3)$ representation across the lattice. The combined statement of this paper and the matter paper, then: defect geometry forces the gauge group

and representation of the strong sector ($\mathfrak{su}(3)$ in the $\mathbf{3}$); the Wilson gauge action propagates that local structure across the D_4 lattice; the dynamical scales of QCD are set by the bond energetics.

References

- [1] T. C. Hales, *A proof of the Kepler conjecture*, Ann. Math. **162**, 1065 (2005), [doi:10.4007/annals.2005.162.1065](https://doi.org/10.4007/annals.2005.162.1065).
- [2] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer (1999), [doi:10.1007/978-1-4757-6568-7](https://doi.org/10.1007/978-1-4757-6568-7).
- [3] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover (1973).
- [4] K. G. Wilson, *Confinement of quarks*, Phys. Rev. D **10**, 2445 (1974), [doi:10.1103/PhysRevD.10.2445](https://doi.org/10.1103/PhysRevD.10.2445).
- [5] K. Symanzik, *Continuum limit and improved action in lattice theories*, Nucl. Phys. B **226**, 187 (1983), [doi:10.1016/0550-3213\(83\)90468-6](https://doi.org/10.1016/0550-3213(83)90468-6).
- [6] G. P. Lepage and P. B. Mackenzie, *On the viability of lattice perturbation theory*, Phys. Rev. D **48**, 2250 (1993), [doi:10.1103/PhysRevD.48.2250](https://doi.org/10.1103/PhysRevD.48.2250).
- [7] W. Celmaster, *Gauge theories on the body-centered hypercubic lattice*, Phys. Rev. D **26**, 2955 (1982), [doi:10.1103/PhysRevD.26.2955](https://doi.org/10.1103/PhysRevD.26.2955).
- [8] D. B. Kaplan, *A method for simulating chiral fermions on the lattice*, Phys. Lett. B **288**, 342 (1992), [doi:10.1016/0370-2693\(92\)91112-M](https://doi.org/10.1016/0370-2693(92)91112-M), [arXiv:hep-lat/9206013](https://arxiv.org/abs/hep-lat/9206013).
- [9] H. B. Nielsen and M. Ninomiya, *Absence of neutrinos on a lattice. I. Proof by homotopy theory*, Nucl. Phys. B **185**, 20 (1981), [doi:10.1016/0550-3213\(81\)90361-8](https://doi.org/10.1016/0550-3213(81)90361-8).
- [10] H. B. Nielsen and M. Ninomiya, *No go theorem for regularizing chiral fermions*, Phys. Lett. B **105**, 219 (1981), [doi:10.1016/0370-2693\(81\)91026-1](https://doi.org/10.1016/0370-2693(81)91026-1).
- [11] R. Kulkarni, *A 67%-Rate CSS Code on the FCC Lattice: $[[192,130,3]]$ from Weight-12 Stabilizers*, [arXiv:2603.20294](https://arxiv.org/abs/2603.20294) [quant-ph] (2026).
- [12] R. Kulkarni, *Matter as Incomplete Crystallization: Quark Charges, Color Confinement, and the Proton Mass from a Single Extra Node in the Vacuum Lattice*, Phys. Open **27**, 100423 (2026), [doi:10.1016/j.physo.2026.100423](https://doi.org/10.1016/j.physo.2026.100423).
- [13] R. Kulkarni, *The mass-energy-information equivalence: A bottom-up identification of the particle spectrum via FCC lattice error correction*, Phys. Open **27**, 100414 (2026), [doi:10.1016/j.physo.2026.100414](https://doi.org/10.1016/j.physo.2026.100414).
- [14] R. Kulkarni, *$\mathfrak{su}(3)$ from a Trapped Tetrahedral Defect in the FCC Lattice*, preprint, SSMTheory Group, IDrive Inc. (2026), [doi:10.5281/zenodo.18503168](https://doi.org/10.5281/zenodo.18503168).
- [15] J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Graduate Texts in Mathematics, vol. 9, Springer (1972), [doi:10.1007/978-1-4612-6398-2](https://doi.org/10.1007/978-1-4612-6398-2).
- [16] Verification script: https://github.com/raghu91302/ssmtheory/blob/main/verify_d4.py.